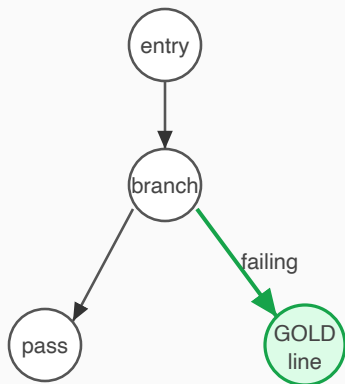


Why execution helps some faults and not others: control-flow signals vs. data-level faults

Coverage / traceback / SBFL record WHICH lines ran. They locate the fault only when the fault changes the control flow.

(a) Control-flow-distinct fault

Failing input drives a DIFFERENT path; gold line is on it.



Failing-only lines
appear in coverage.

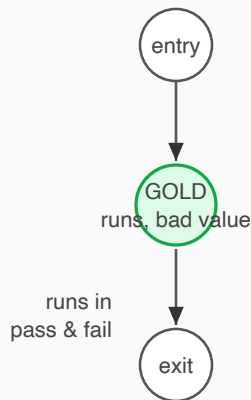
- ✓ coverage filter helps
- ✓ SBFL differential $\neq \emptyset$
- ✓ traceback points near

Example: matplotlib-22711

93 lines run only under failure; 5 / 13 gold lines lie in that failing-only set.

(b) Data-level fault

Same path runs; wrong VALUE, or fix is on a skipped branch.



Gold also runs in passing
tests → no differential.

- ✗ SBFL differential = $\emptyset$
- ✗ no ranking signal
(only membership = filter)

Examples: astropy-14182 (0/14 gold in

failing-only) · django-14017 (fix on a missed branch the buggy run never took).